

Avancee en apprentissage statistique

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September 7, 2026

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1 Monday September 7th

I arrived to this class and also missed last week's class. They talked about parameter estimation and maximum likelihood, it seems. We are now moving onto multiple linear regression.

1.1 Linear Regression

Simple linear regression is a special case of multiple linear regression. Consider the case in which we have a scalar real-valued output variable $y \in \mathbb{R}$ and a vector of input variables $\mathbf{x} \in \mathbb{R}^n$. In linear regression, we have a vector of weights $\mathbf{w} \in \mathbb{R}^n$ and a zero-mean gaussian random variable $\varepsilon \sim N(0, P^2)$. We declare $f(x)$, a function that relates our weights to a predicted value of y .

$$f(x) = \mathbf{w}^\top \mathbf{x} \quad (1)$$

with this function, we can measure the difference between our predictions and our real-value and attribute it to the error ε .

$$y = f(x) + \varepsilon \quad (2)$$

$$\hat{y} \approx f(x) \quad (3)$$

Naturally, this leads to the question of how we determine our weight vector $\mathbf{w}$ that we use to get our linear regression model. If we have a data sample: $\{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$, we can use that to get our best guess for those values of the weight vector. We can use maximum likelihood estimate to estimate the values.

$$L(w|data) = \mathbb{P}(data|w) = \prod_{i=1}^m g_y(y_i|x_i, w) \quad (4)$$

$$= \prod_{i=1}^m (2\pi)^{-1/2} (P^2)^{-1/2} e^{-\frac{1}{2P^2} (y_i - \mathbf{w}^\top \mathbf{x})^2} \quad (5)$$

$$= \prod_{i=1}^m (2\pi)^{-1/2} (P^2)^{-1/2} e^{-\frac{1}{2P^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2} \quad \text{matrix case} \quad (6)$$

We can stack our data samples next to each other in vectors to facilitate this maximum likelihood.

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}, \mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_m^\top \end{bmatrix}, \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \quad (7)$$

So we have $\mathbf{y} \in \mathbb{R}^m, \mathbf{X} \in \mathbb{R}^{m \times n}, \mathbf{w} \in \mathbb{R}^n$. With maximum likelihood estimation, we can solve the below least-squares equation to get our predicted $\hat{\mathbf{w}}$.

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \mathbf{X}^\dagger \mathbf{y} \quad (8)$$

We can then get our predicted values for y , based off of any x using the model $f(x) = \mathbf{w}^\top \mathbf{x}$. If we have a series of inputs $\mathbf{X}$, we can turn it into a series of predictions with

$$\hat{\mathbf{y}} = \mathbf{X} \hat{\mathbf{w}} \quad (9)$$

What is the distribution of $\hat{\mathbf{w}}$?

$$\hat{\mathbf{w}} \sim N(w, P^2 (\mathbf{X}^\top \mathbf{X})^{-1}) \quad (10)$$

We will prove this in the practical session.

We can use the maximum likelihood estimate to estimate our value for P

$$\hat{P}_{ML}^2 = \frac{1}{m} \sum_{i=1}^m \hat{\varepsilon}_i^2 \quad (11)$$

where $\hat{\varepsilon} = y_i - \hat{y}_i$. It is an estimate because we are using the predicted weights $\hat{\mathbf{w}}$. We then remove the bias by doing the empirical

$$\hat{P}^2 = \frac{1}{m - (n - 1)} \sum_{i=1}^m \hat{\varepsilon}_i^2 \quad (12)$$

Remark: We might have problems when computing $(\mathbf{X}^\top \mathbf{X})^{-1}$. If we have repeated data points, then the matrix is not full rank and it is therefore not invertible. We can add a tiny noise onto the diagonal of the matrix so that we ensure full rank, and that we can compute the inverse.

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X} + \xi I_p)^{-1} \mathbf{X}^\top \mathbf{y} \quad (13)$$

where ξ is a tiny noise with arbitrary distribution.

1.2 Bayesian LR

Here we move onto a bayesian form of linear regression. I do not fully understand what that is at the moment.

We set a prior on $w \sim N(0, \Sigma_n)$, that it follows a normal distribution with zero mean and covariance Σ_p .

$$P(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \frac{P_y(y|\mathbf{X}, \mathbf{w})P(\mathbf{w})}{P_y(\mathbf{y}|\mathbf{X})} \quad (14)$$

$$P(\mathbf{w}|\mathbf{X}, \mathbf{y}) \propto e^{-\frac{1}{2}((\mathbf{w}-\bar{\mathbf{w}})^\top \mathbf{A}(\mathbf{w}-\bar{\mathbf{w}}))} \quad (15)$$

where

$$\bar{\mathbf{w}} = \sigma^{-2} \mathbf{A} \mathbf{X}^\top \mathbf{y}, \quad \mathbf{A} = \sigma^{-2} \mathbf{X}^\top \mathbf{X} + \Sigma_n^{-1} \quad (16)$$

If we don't put any weight on our covariance matrix Σ , then we end up getting the same maximum likelihood estimation as the $\hat{\mathbf{w}}$ ends up being the same as previously. If we have it be the ξI_n , then it is the same as the previous with that tiny regularization that allows us to compute the inverse. We can choose whatever we want (I am still unclear on how or why we would do that) for Σ_n , as long as it is a covariance matrix—symmetric ($\Sigma = \Sigma^\top$) and PSD ($\Sigma \succeq 0$).

1.3 Predictions

Why would we want other covariances? It seems that it enables us to regularize or penalize the model by putting assumptions on the covariance. It allows us as the model designer to have some say in how good we think our model is, and then embed those assumptions in the model to improve them accordingly. Maximum Likelihood goes to Maximum A Posteriori.

$$\hat{y}_{MAP} = \mathbf{x}^\top \hat{\mathbf{w}}_{MAP} \quad (17)$$

This allows us to have a posterior $\mathbf{w} \sim N(\bar{\mathbf{w}}, \mathbf{A}^{-1})$. It helps us get a confidence on our outputs because the covariance matrix gives us the full predictive distribution. I can't pretend I fully understand what that means yet but I am much closer.

1.3.1 Predictive Distribution

We now have the same training data set $\mathbf{X} \in \mathbb{R}^{m \times n}, \mathbf{y} \in \mathbb{R}^m$. Now, we get new inputs $\mathbf{X}^*$, and we want to not only have an output but we also want to have a confidence of our prediction based off of the inputs and the output we get f^* .

Our predictive maximum a posteriori distribution propagates into the distribution of our vector of new outputs. So we can model a portion of our uncertainty that comes from our weights.